PER SCHOLAS

Guided Lab 404.5.2 - From Excel Workbook to Stunning Report and Dashboard in Power BI Desktop

Lab Overview:

This guided lab walks you through the process of transforming data from an Excel workbook into a comprehensive report using Power BI Desktop. You will learn to import, clean, and transform data; create DAX expressions for new measures and tables; establish relationships between tables; and build interactive visualizations. The lab culminates in creating a finished executive summary report that answers key business questions. You will also learn basic formatting techniques to enhance the report's presentation.

Learning Objectives:

- Explain the purpose of transforming data before loading it into Power BI.
- Describe how Power BI detects numeric and date fields.
- Illustrate how relationships between tables are created and their importance.
- Import the Excel workbook into Power BI Desktop.
- Transform data types, format columns, and filter data using Power Query Editor.
- Write DAX expressions to create a new measure.
- Create a relationship between the tables.
- Build a report with various visualizations, including line charts, maps, and clustered column charts.
- Use slicers to filter report data.
- Analyze the impact of data transformations on the final report.
- Examine how relationships between tables affect data analysis.
- Determine which segments and products are performing best based on the report data.
- Judge the usefulness of slicers for interactive data exploration.
- Design a comprehensive executive summary report in Power BI Desktop.

Download the sample.

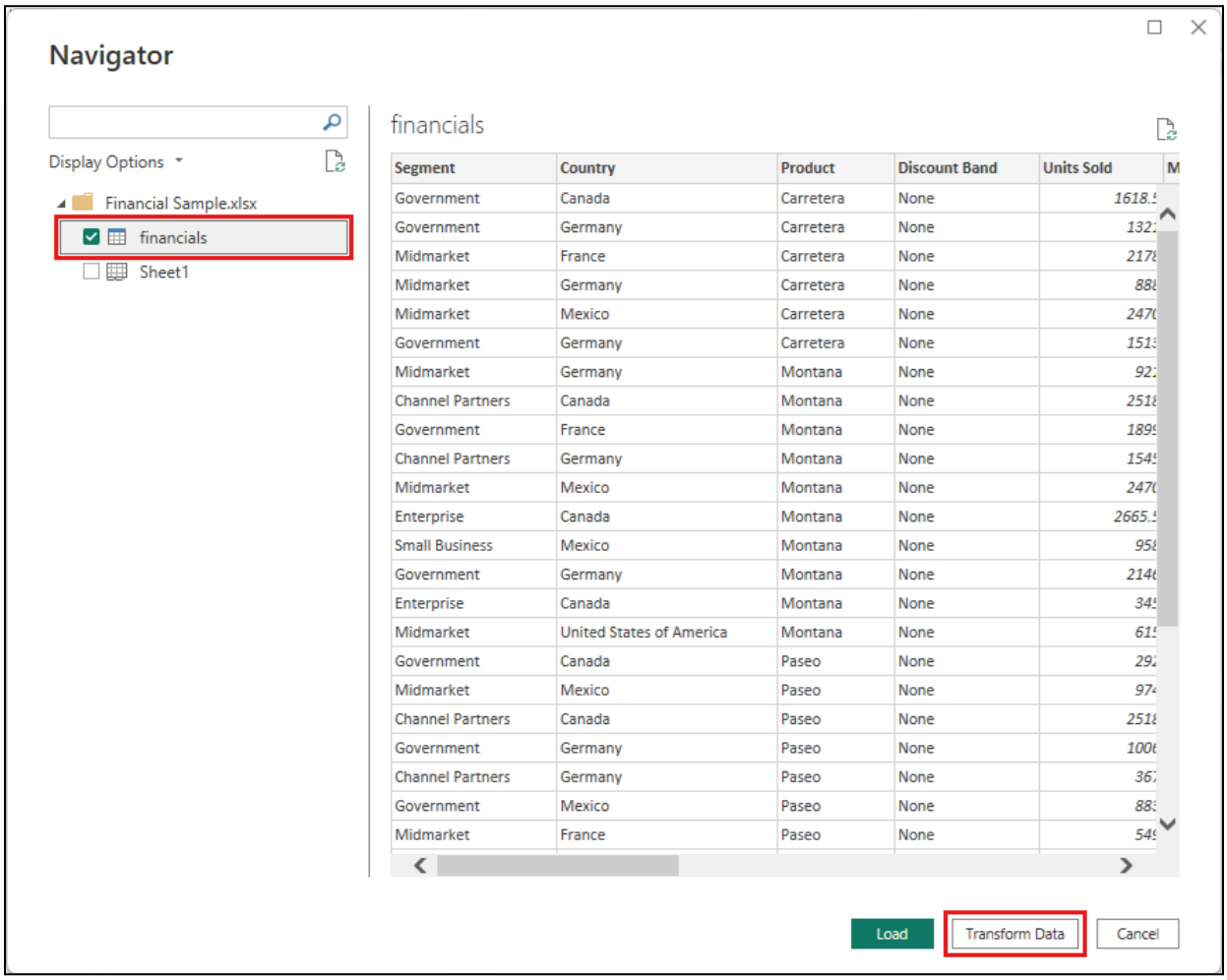
1. Click here to download the Financial Sample Excel workbook.
2. Open Power BI Desktop.
3. On the Home page, select Excel workbook.
4. Navigate to where you saved the sample workbook and select Open.

Begin Instructions:

Step 1: Prepare your data.

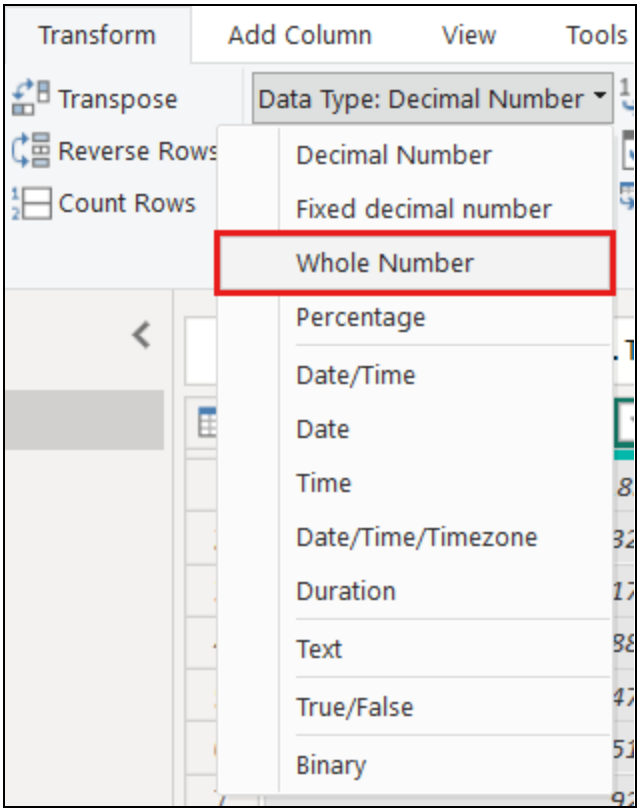
In Navigator, you have the option to **transform** or **load** the data. The Navigator provides a preview of your data so you can verify that you have the correct range of data. Numeric data types are italicized. If you need to make changes, transform your data before loading. To make the visualizations easier to read later, we want to transform the data for this exercise. As you make each transformation in Power Query Editor, you see it added to the list under Query Settings in APPLIED STEPS.

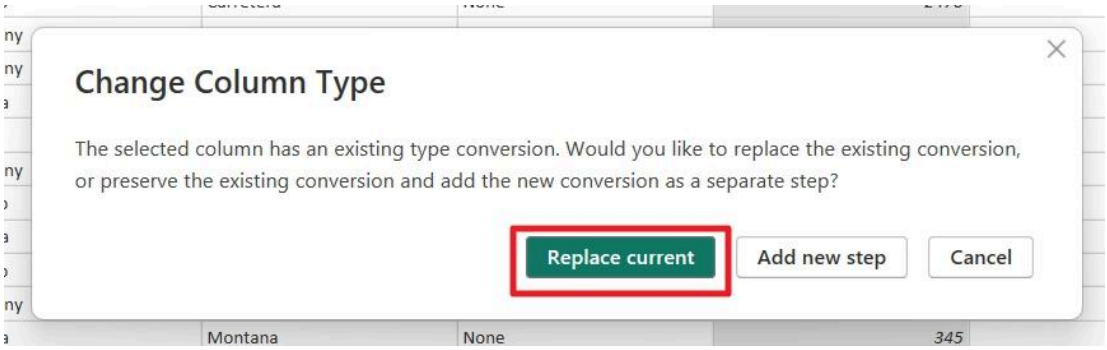
1. Select the **checkbox** for the `**financials**` table, and then select the **Transform Data** button.



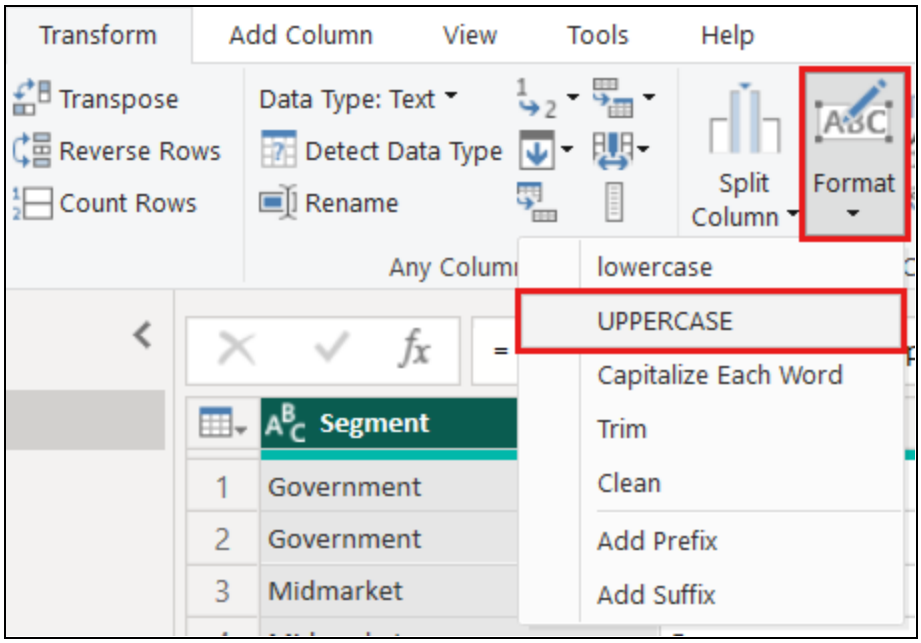
2. The **Power Query Editor** opens. Select the `**Units Sold**` column. On the **Transform** tab, select **Data Type**, and then select **Whole Number**. In the Change Column Type dialog box that opens, choose **Replace current.** As shown in the screenshot below.

Changing data types is the data cleaning step that users do most often. In this case, the units sold are in decimal form. It does not make sense to have 0.2 or 0.5 of a unit sold, so we changed it to a whole number.

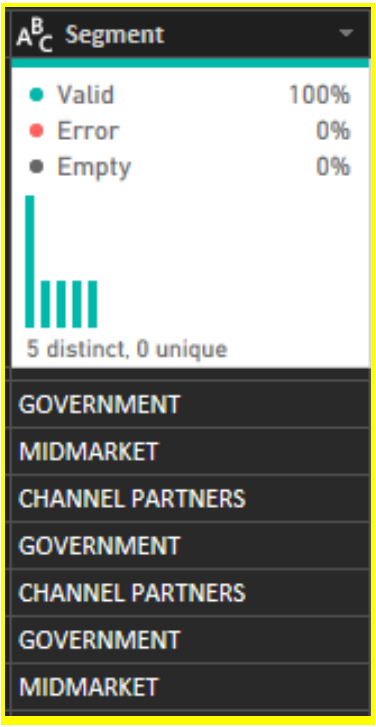




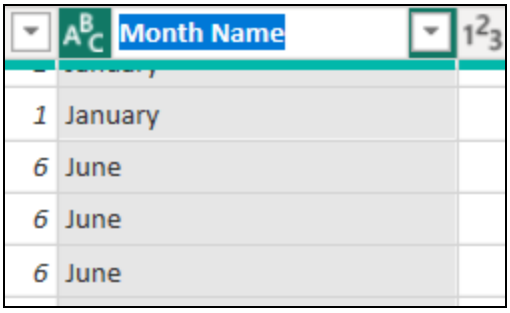
3. We want to make the segments easier to see in the chart later, so let's format the data in the Segment column. Select the `**Segment**` column. On the **Transform** tab, select **Format**, and then select **UPPERCASE**.



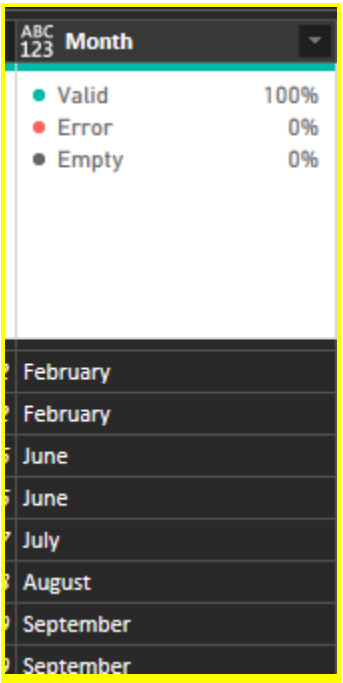
MY SCREENSHOT



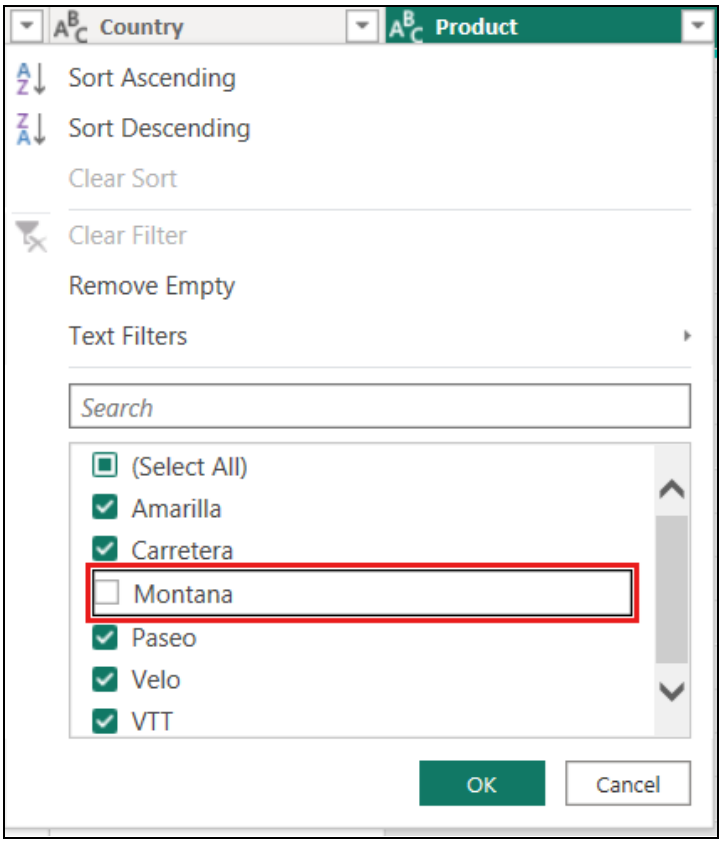
4. Let's shorten the column name from **Month Name** to just Month. Double-click on the **Month Name** column, and rename it to **Month**.



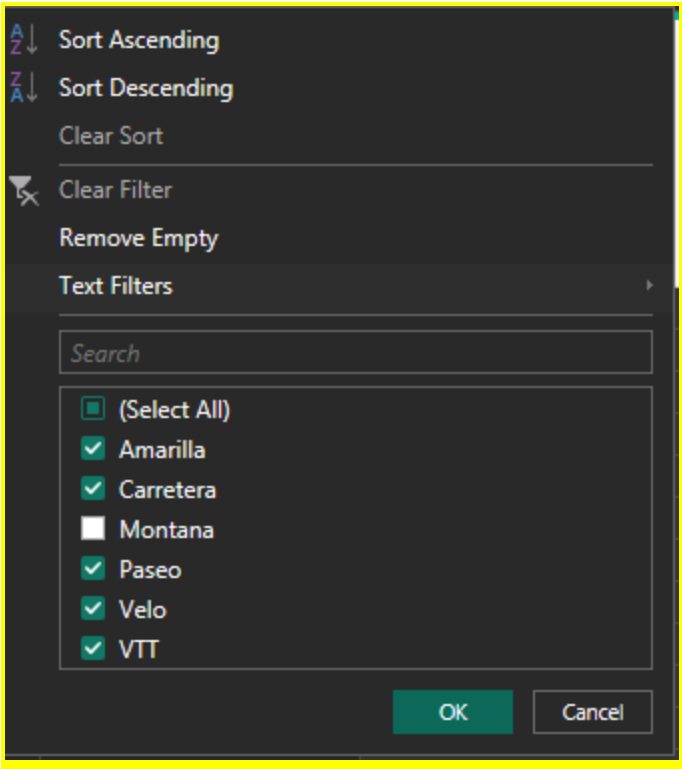
MY SCREENSHOT



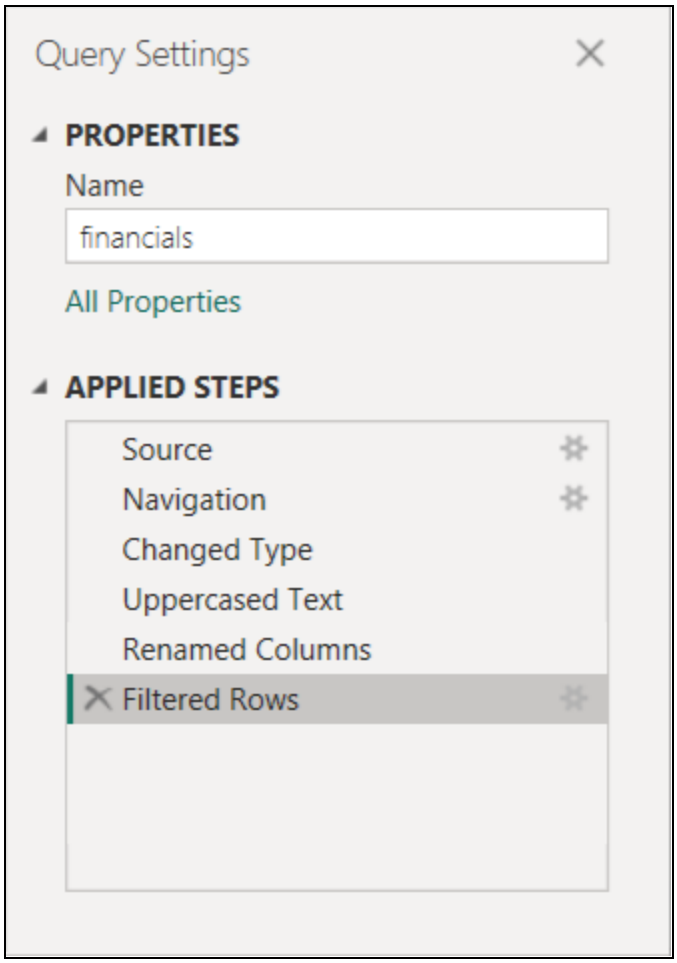
5. On the **Product** column header, select the **dropdown menu** and clear the box next to **Montana**. We know that the **Montana** product was discontinued last month, so we want to filter this data from our report to avoid confusion. Click the **OK** button.



MY SCREENSHOT

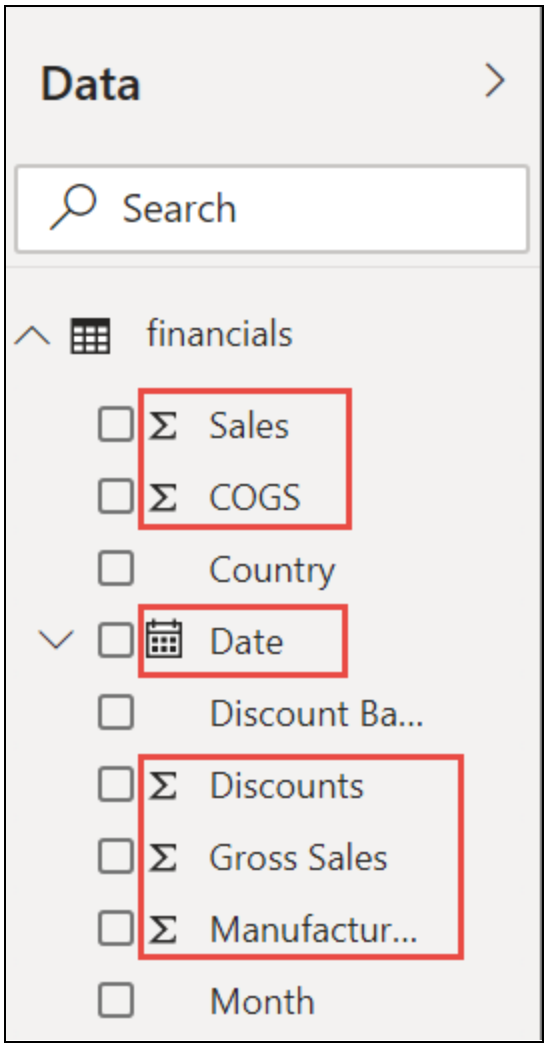


6. You can see that each transformation has been added to the list under Query Settings in **APPLIED STEPS.**

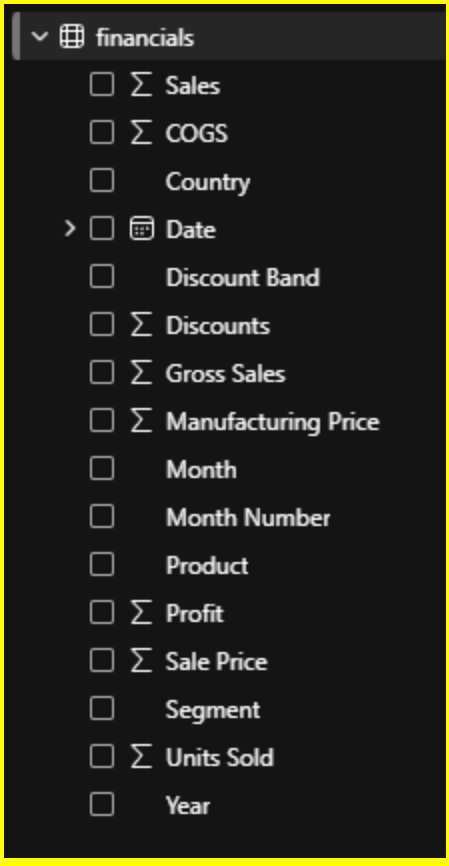


7. Go to the **Home** tab and Select **Close & Apply**. Our data is almost ready for building a report.

The sigma symbol in the Data list indicates a field that Power BI has detected is numeric. Power BI uses a calendar symbol to denote a date field.



MY SCREENSHOT

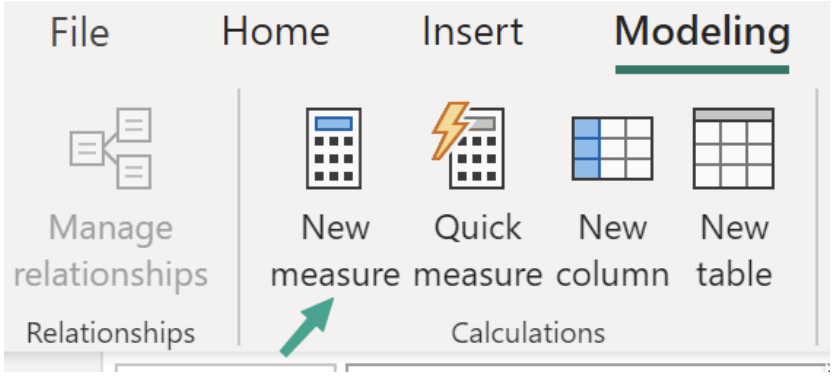


Step 2: Write expressions or formulas in DAX.

Writing ***measures*** and creating ***tables*** in the **DAX** formula language is super powerful for data modeling. There is much to learn about DAX in the Power BI documentation. For now, let's write a basic expression to create a **new measure**, and then write another expression to create a whole table.

2.2: Create a new measure column.

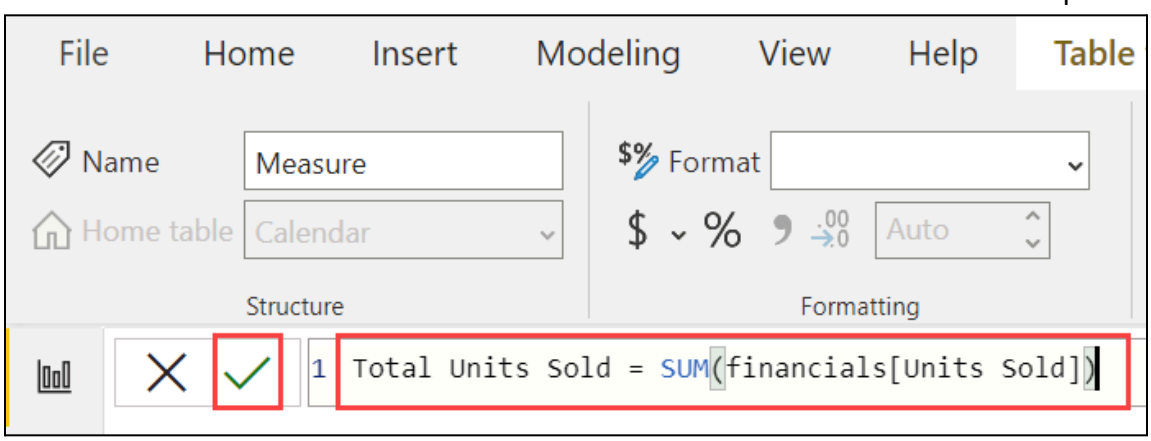
2.2.1. Navigate to the **"Modeling"** tab in the top ribbon and select **"New measure."**



2.2.2. Enter the following DAX formula in the formula bar at the top of the report canvas to add all of the numbers in the Units Sold column.

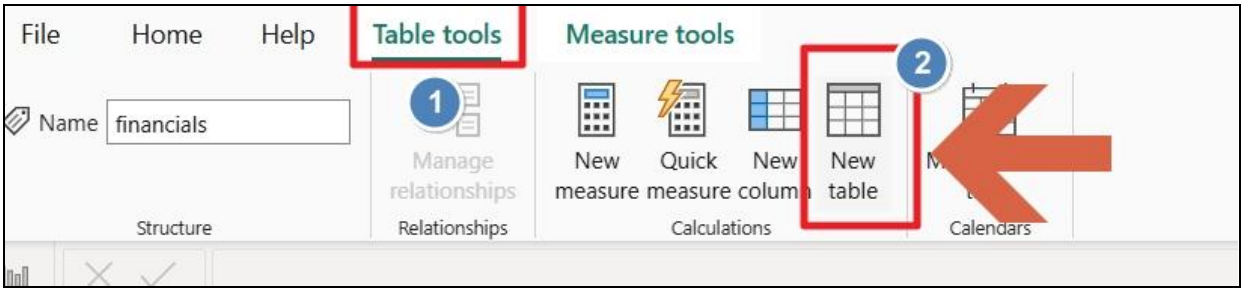
```
None
Total Units Sold = SUM(financials[Units Sold])
```

2.2.3. Select the check mark to commit. The measure is added to the list in the Data pane.



2.3: Create a new table.

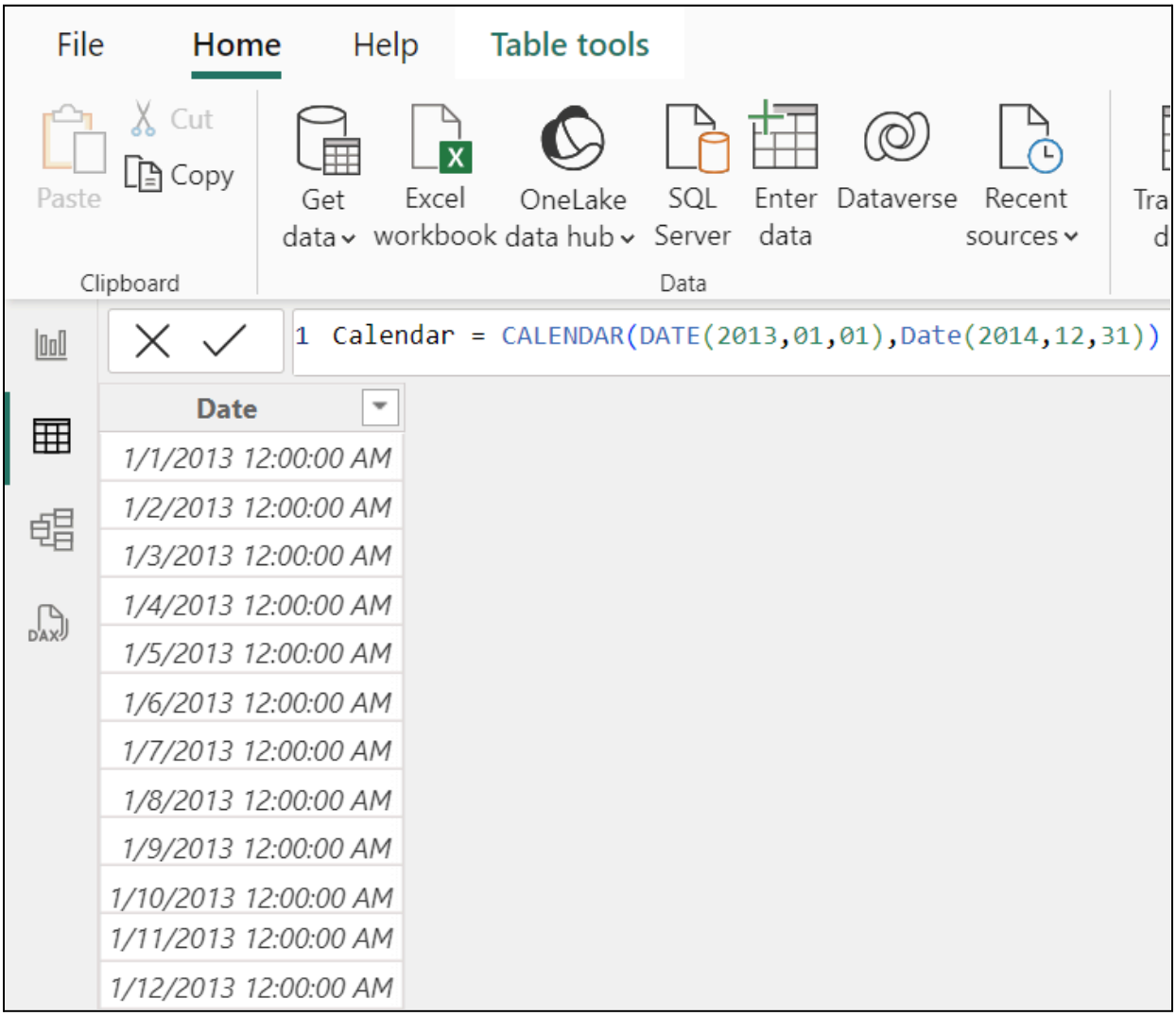
2.3.1. Navigate to the **"Table tools"** tab in the top ribbon and select **"New table."**



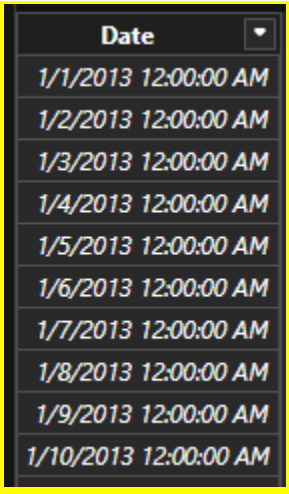
2.3.2. Enter the following DAX formula in the formula bar to generate a Calendar table of all dates between January 1, 2013, and December 31, 2014.

```
Calendar = CALENDAR(DATE(2013,01,01),Date(2014,12,31))
```

2.3.3. Here are the first 23 rows of the single-column table that you just created.



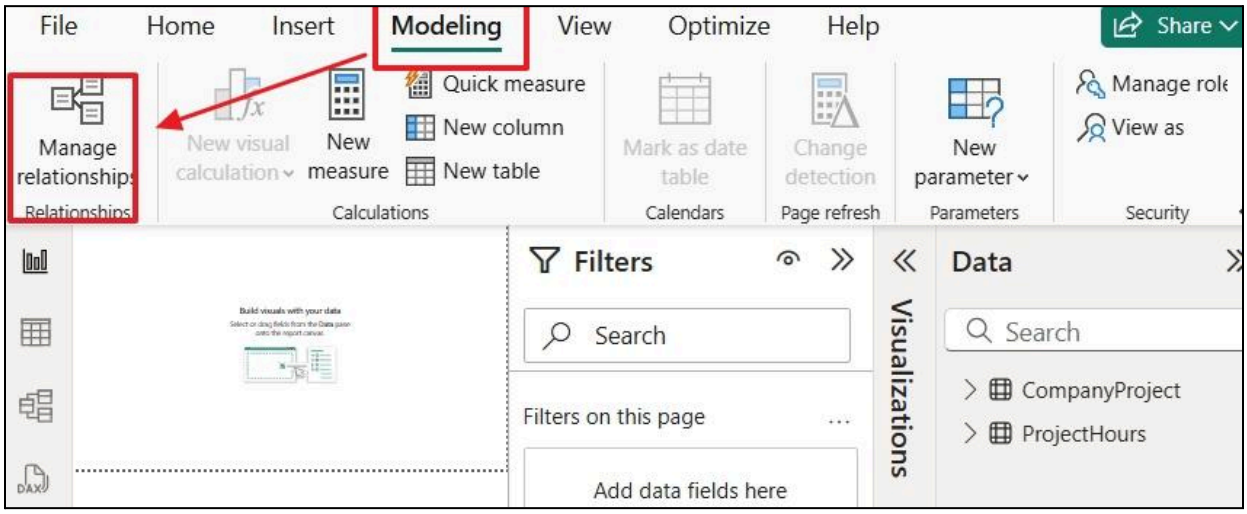
MY SCREENSHOT



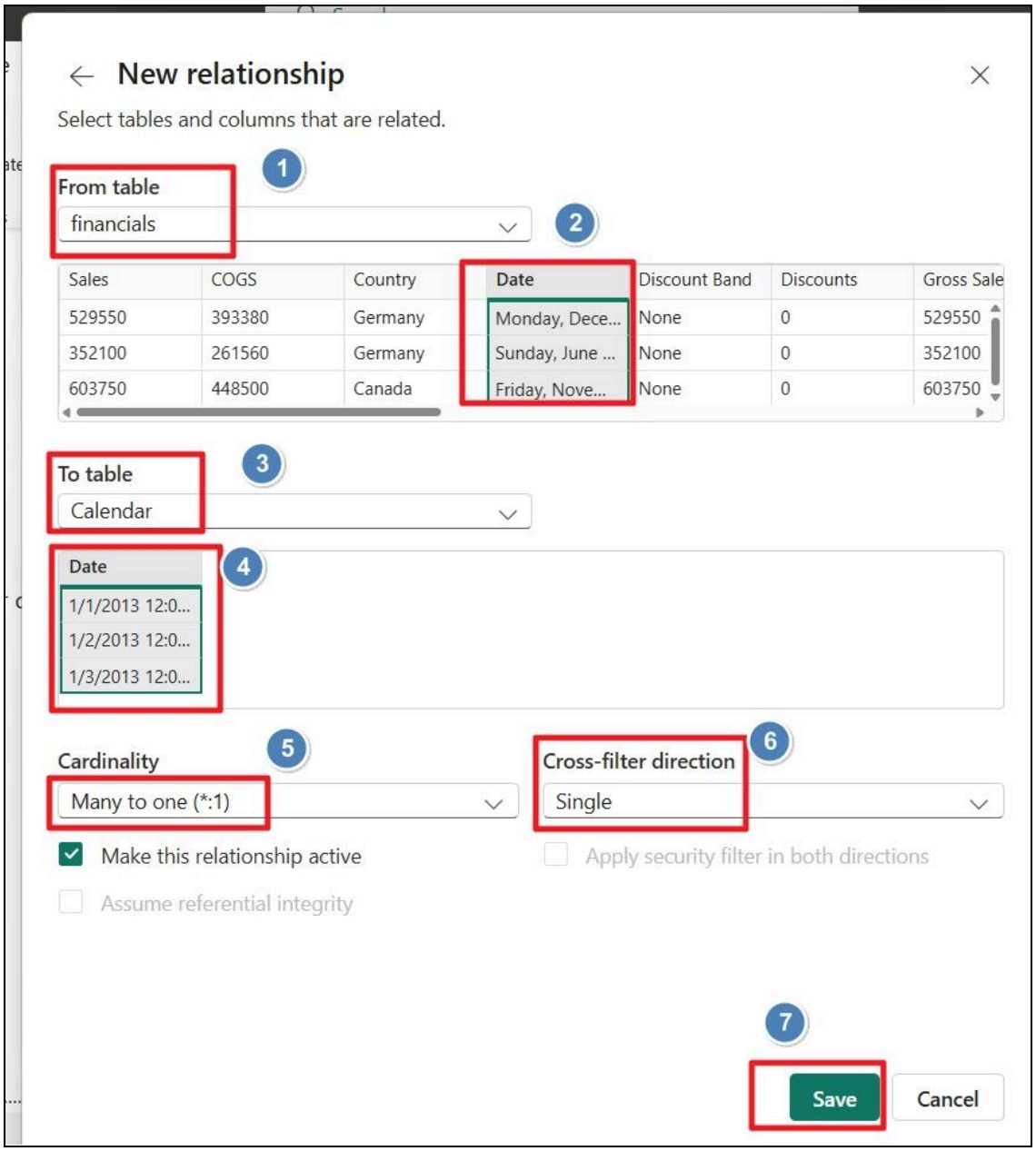
Date
1/1/2013 12:00:00 AM
1/2/2013 12:00:00 AM
1/3/2013 12:00:00 AM
1/4/2013 12:00:00 AM
1/5/2013 12:00:00 AM
1/6/2013 12:00:00 AM
1/7/2013 12:00:00 AM
1/8/2013 12:00:00 AM
1/9/2013 12:00:00 AM
1/10/2013 12:00:00 AM

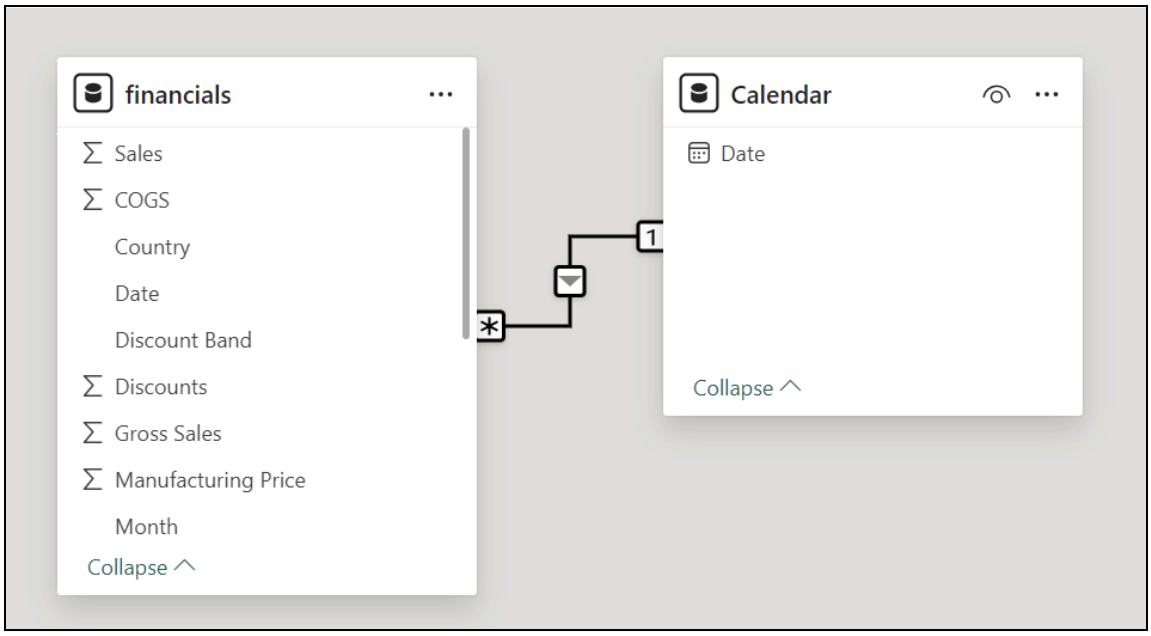
Step 3: Create a relationship between Tables (Data modeling)

1. In the top ribbon, on the **Modeling** tab, select **`Manage relationships`** → **New relationship.**

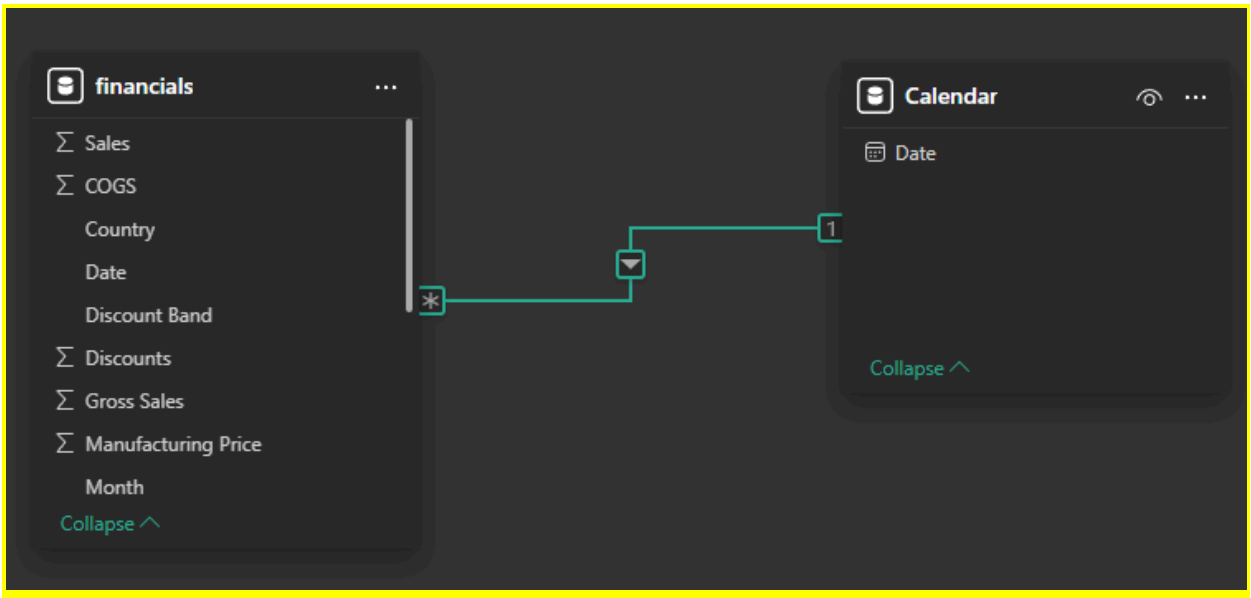


2. The **New Relationship** dialog box will pop up. In the `**From table**` drop-down list, select a `**financials**` table. Then select the column you want to use in the relationship.
    - In our case, select the `**Date**` column.
3. In the `**To table**` drop-down list, select the `**Calender**` table. Select the other column you want to use in the relationship.
    - In our case, select the `**Date**` column.
    - In the **Cardinality**: Select **Many to One**
    - In the **Cross-filter direction:** Select **single**
    - Click on the **Save** button**.**


New relationship
Select tables and columns that are related.
1
From table
financials
2
Sales
COGS
Country
Date
Discount Band
Discounts
Gross Sale
529550
393380
Germany
Monday, Dece...
None
0
529550
352100
261560
Germany
Sunday, June ...
None
0
352100
603750
448500
Canada
Friday, Nove...
None
0
603750
3
To table
Calendar
4
Date
1/1/2013 12:0...
1/2/2013 12:0...
1/3/2013 12:0...
5
Cardinality
Many to one (*:1)
6
Cross-filter direction
Single
Make this relationship active
Apply security filter in both directions
Assume referential integrity
7
Save
Cancel



MY SCREENSHOT

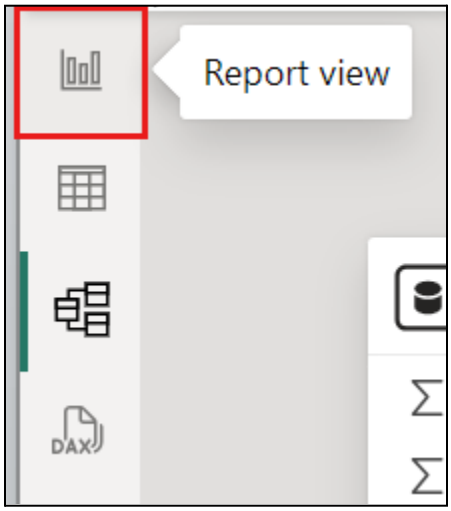


Established a many-to-one cardinality relationship between calendar and financials tables. The dates in Calendar are unique and there's multiple corresponding values in the Financials table.

4. Check on **Financials(Date)**→ Click on the **close** button.

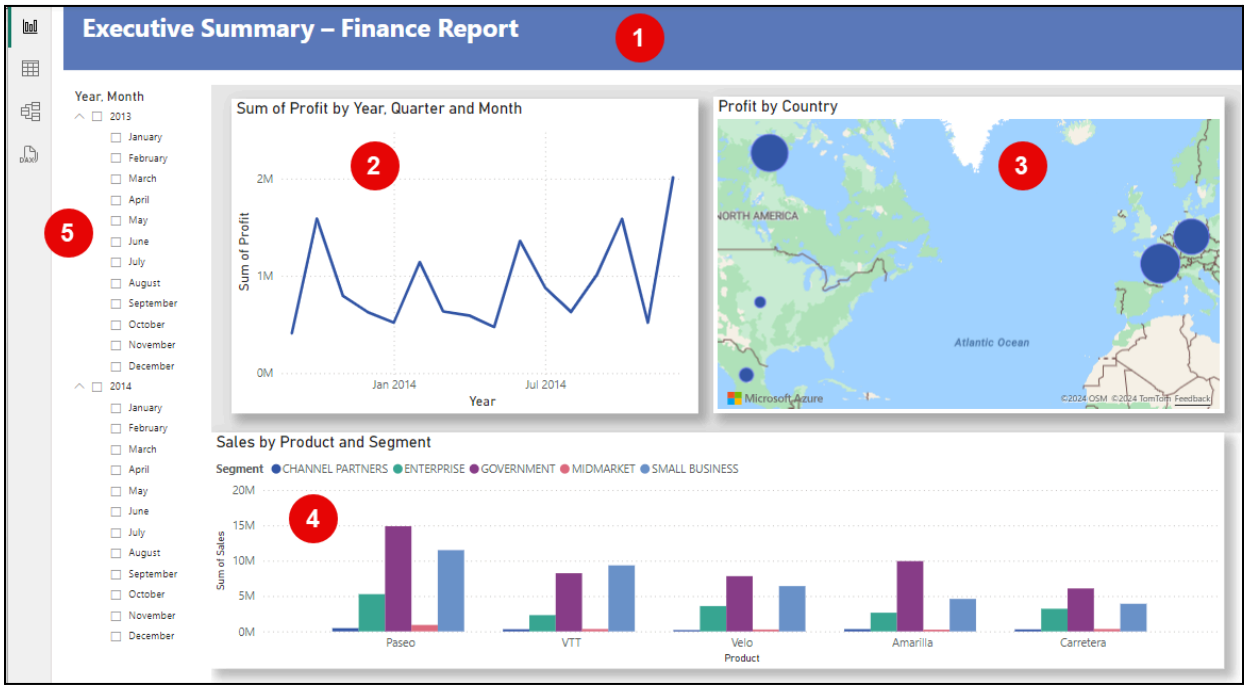
Step 4: Build your report.

Now that you have transformed and loaded your data, it is time to create your report. In the left rail, select the **Report view** icon from the left panel as shown in the screenshot below.



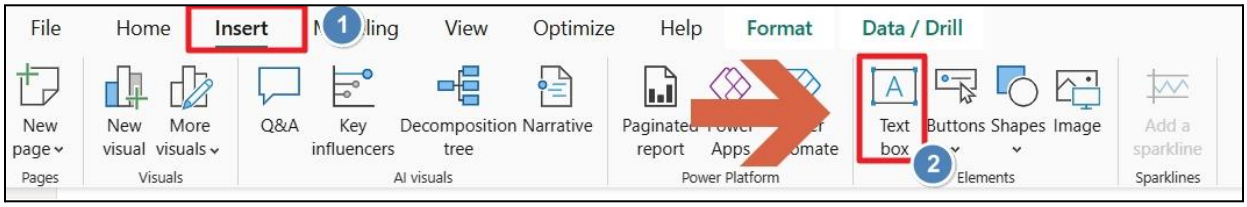
In the **Data** pane on the right, you see the fields in the data model you created.

Let's build the final report, one visual at a time.

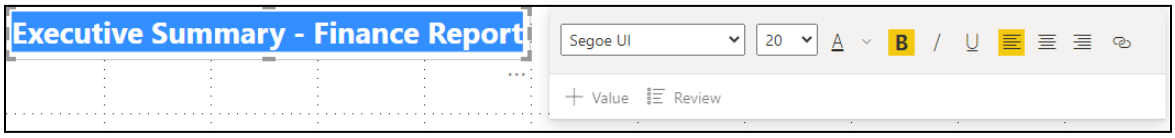


Visual 1: Add a title.

1. On the **Insert** ribbon, select the **Text** box. Inside the text box, type **"Executive Summary – Finance Report."**



2. Select the text you typed. Make the font bold and set the size to 20.



3. Resize the box to fit on one line.

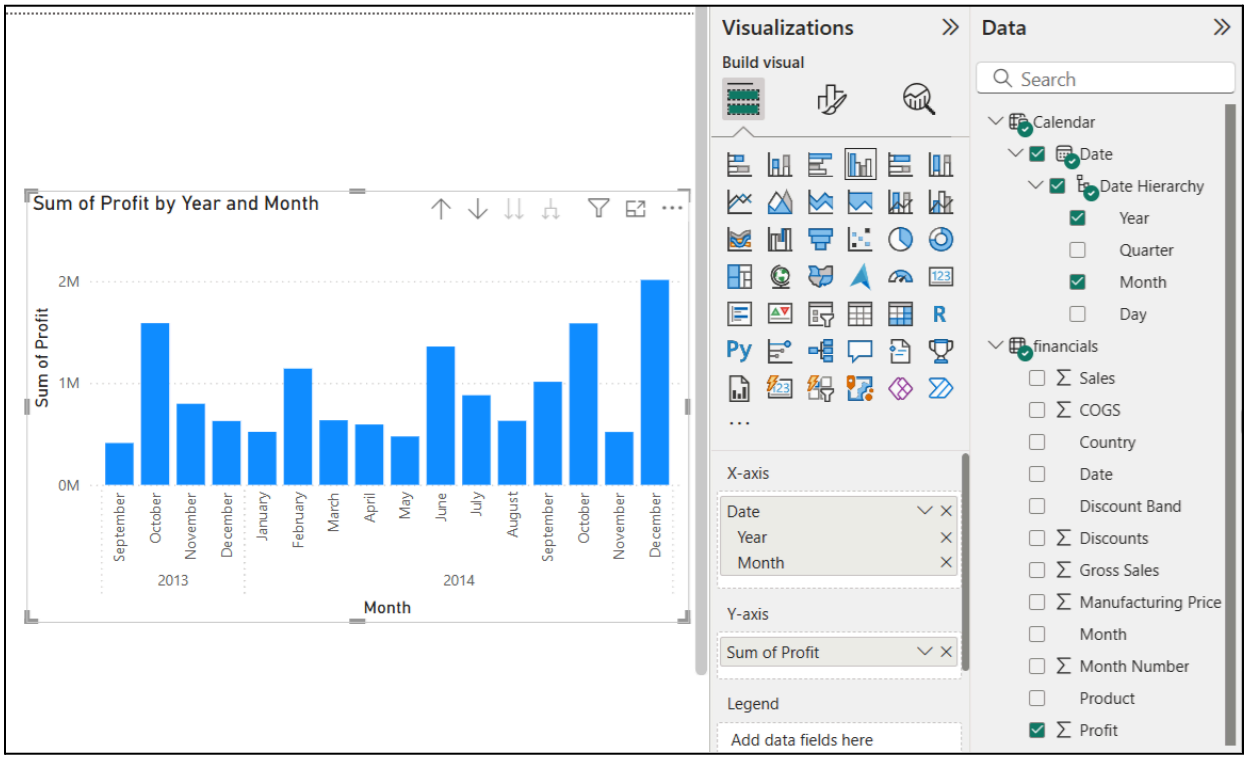
MY SCREENSHOT

Visual 2: Profit by date.

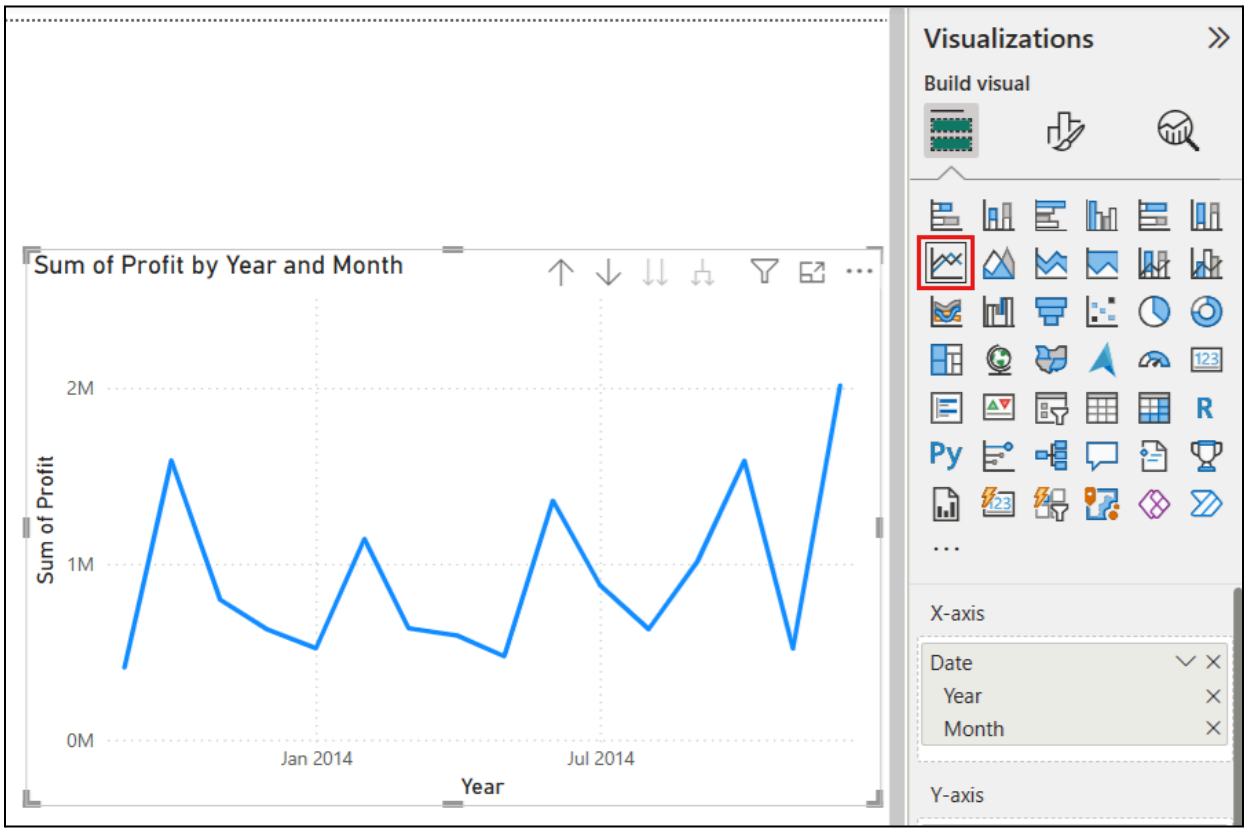
Let's create a line chart to see which month and year had the highest profit.

1. From the **financials** table**,** drag the `**Profit**` field to a blank area on the report canvas. By default, Power BI displays a column chart with one column, Profit.

2. From the **Calendar** table, Drag the `**Date**` field to the same visual and remove the Quarter and **Day** columns**.** Power BI updates the column chart to show monthly profit for the two years.

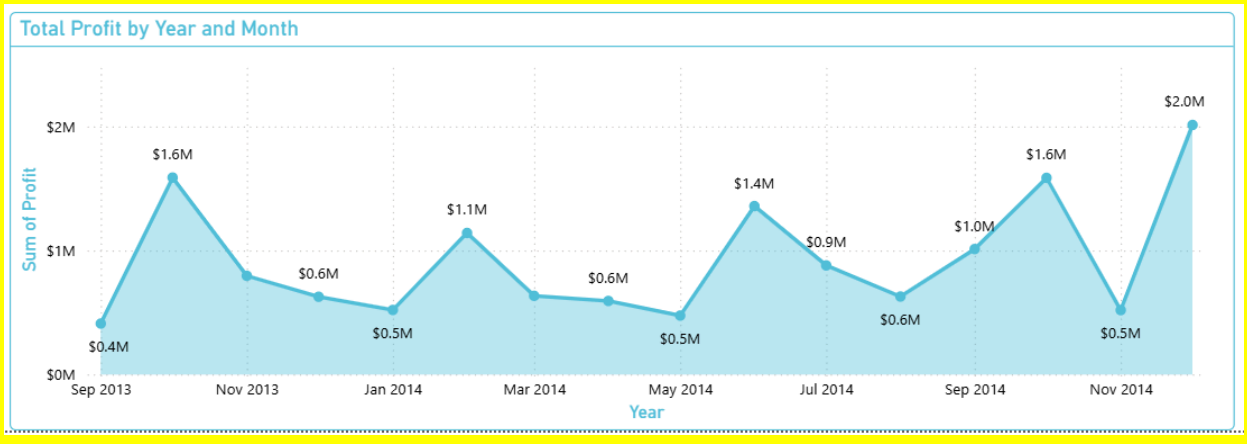


3. On the **Build visual** tab, change the visualization type to **Line chart** as shown in the screenshot below.



Now you can easily see that **December 2014** had the highest profit.

MY SCREENSHOT



For some reason November 2013, March 2014 are not labeled even though all others are. When filtered down to

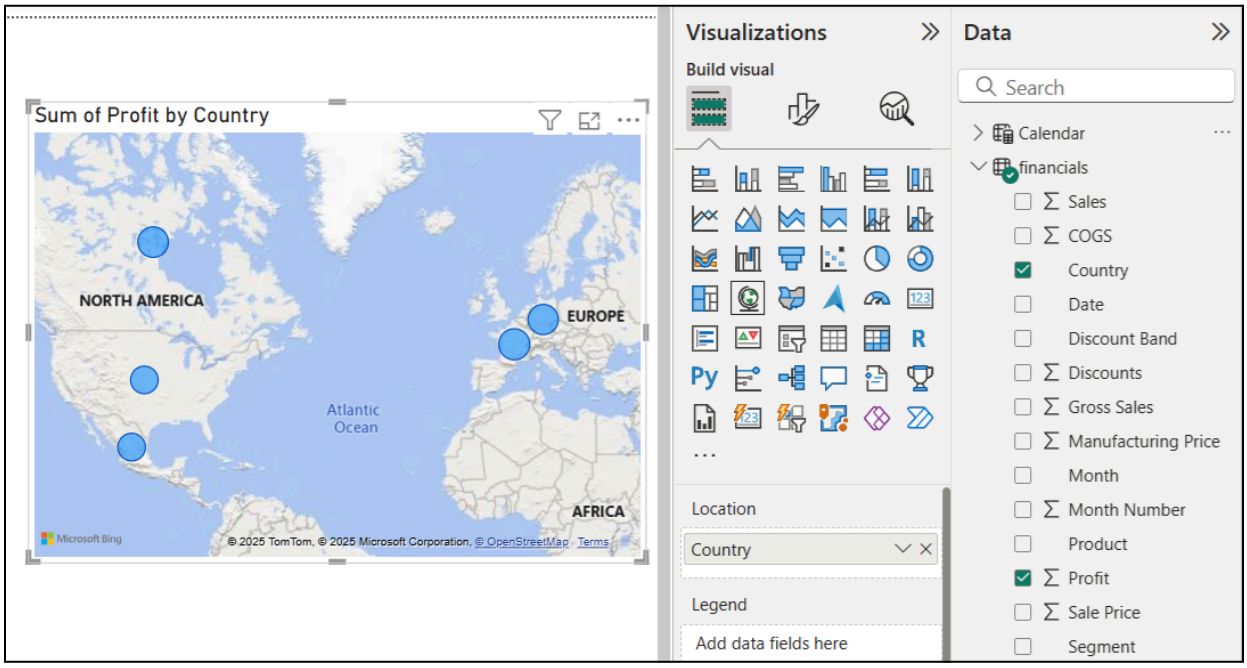
those months, powerbi shows the value, otherwise in full view it does not show.

Visual 3: Profit by country/region.

Let's create a map to see which country/region had the highest profits.

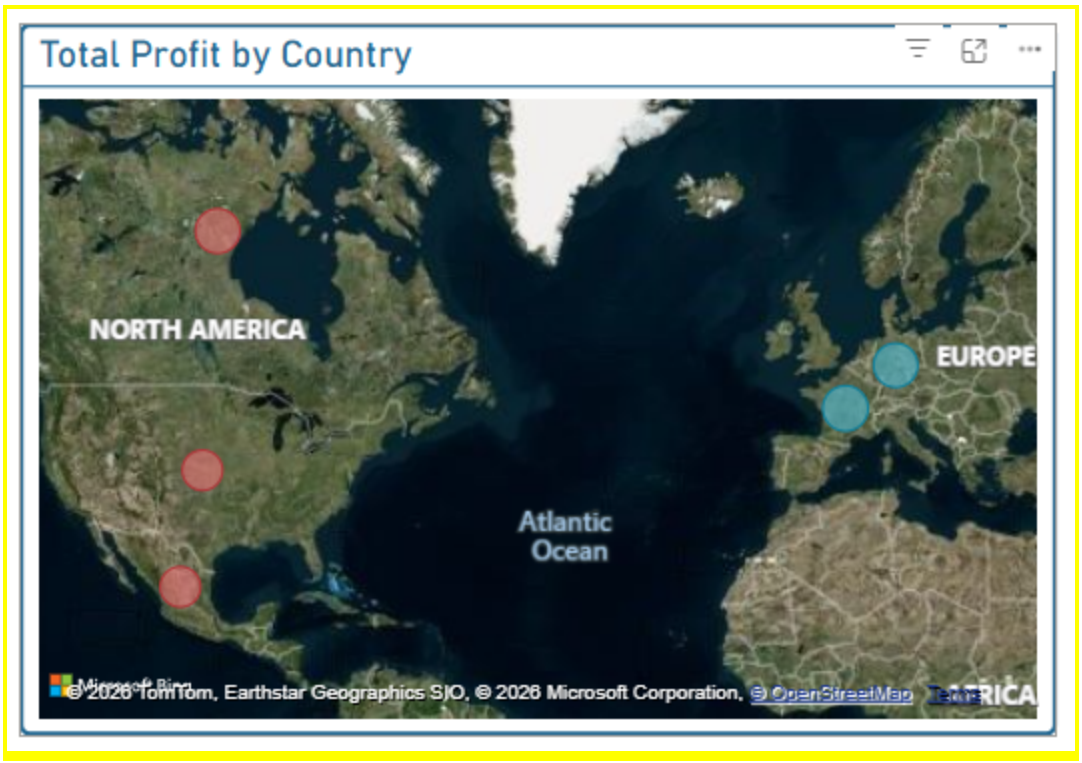
1. From the **financials** table**,** drag and drop the `**Country**` field to a blank area on your report canvas to create a map.
2. From the **financials** table**,** drag and drop the `**Profit**` field to the map.

- Power BI adds bubbles representing the relative profit of each location to the map visual.



Europe seems to be performing better than North America.

MY SCREENSHOT

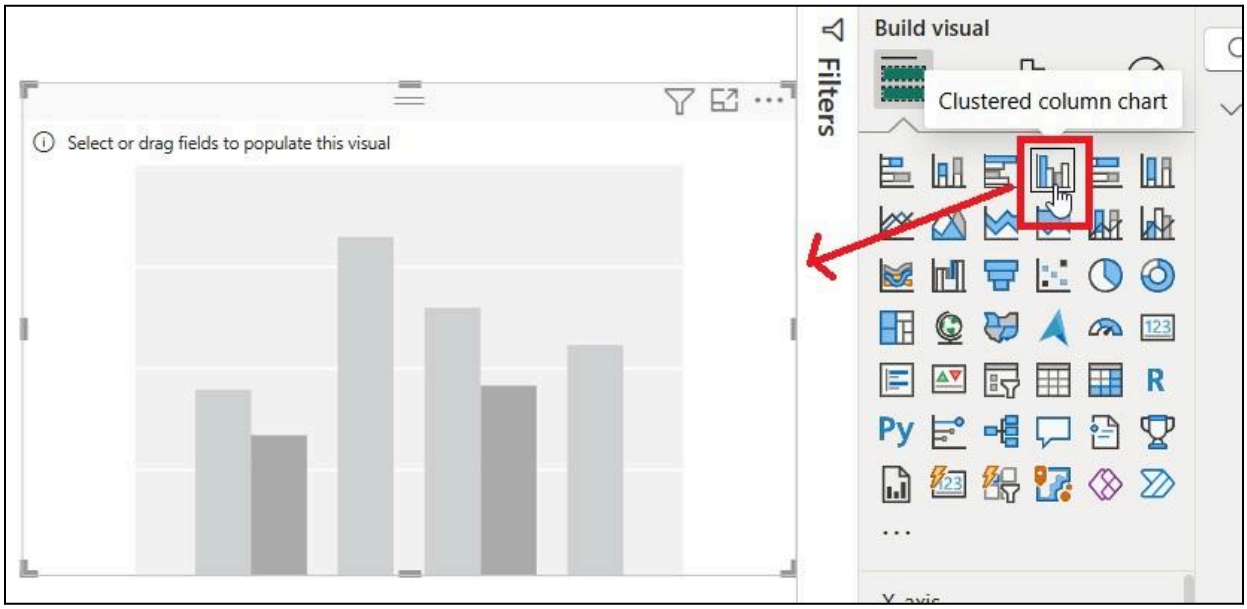


The bubbles are very close in size, so I chose to change the european colors to indicate that they are responsible for higher profit.

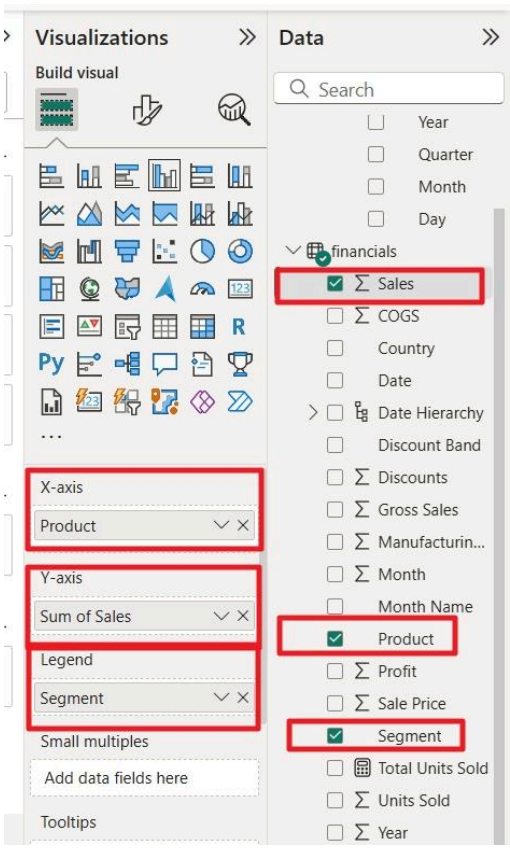
Visual 4: Sales by product and segment.

Let's create a **Clustered Column bar** chart to determine which companies and segments to invest in.

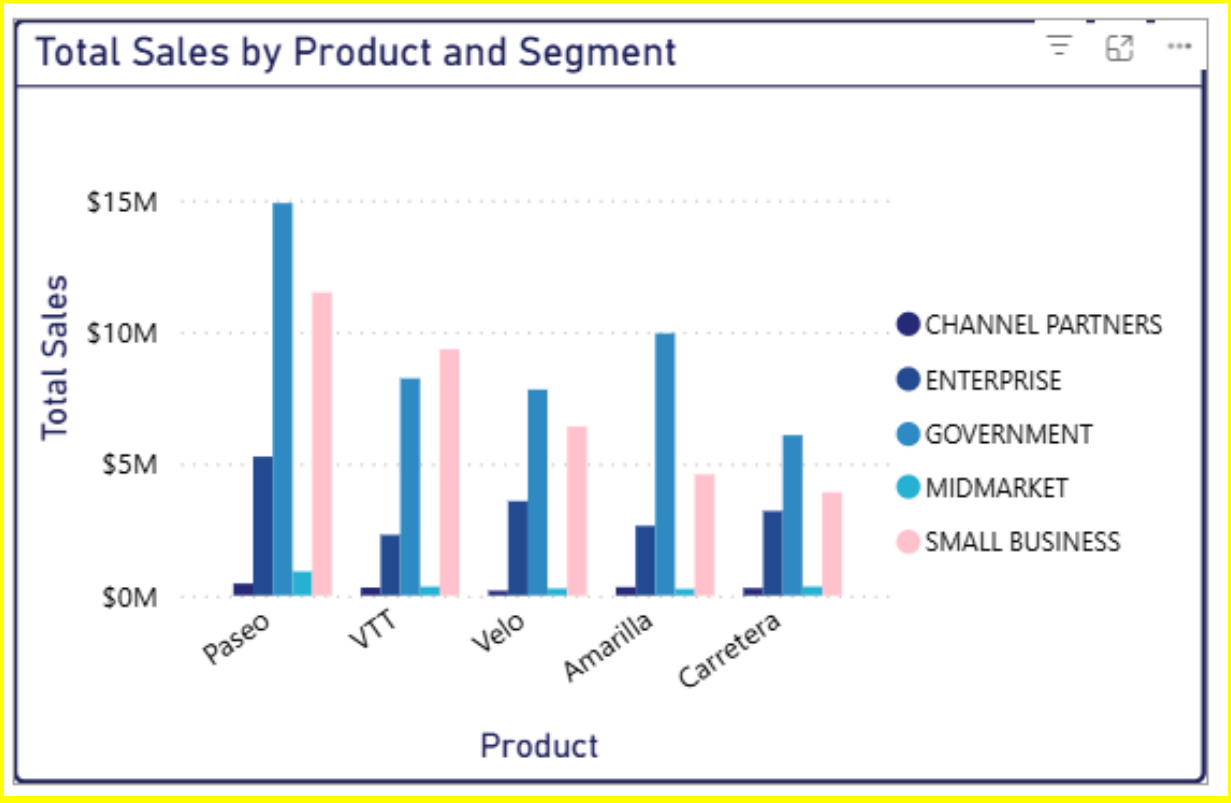
1. Drag the two charts that you created side-by-side in the top half of the canvas. Save some room on the left side of the canvas.
2. From the **Visualizations** pane, click on the `**Clustered column chart**` icon as shown in the screenshot below.



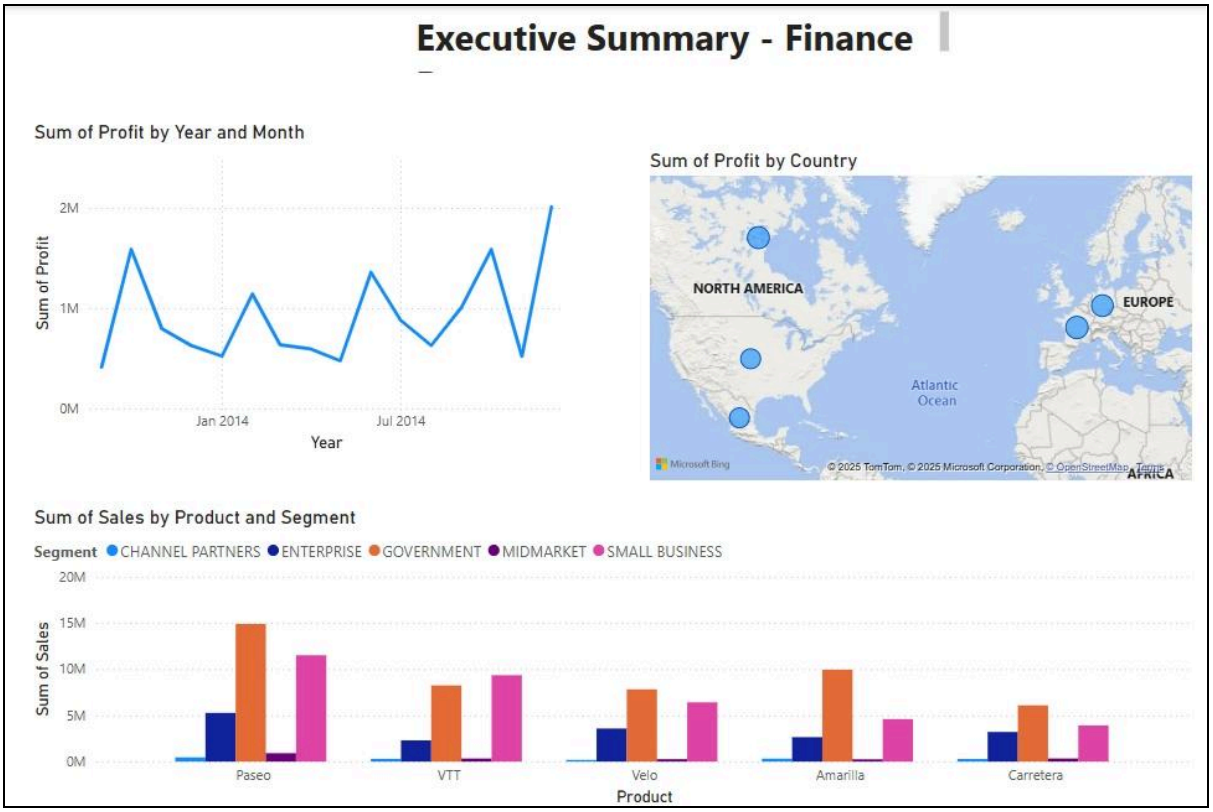
3. From the `**Finanacial**` sheet:
    a. Drag and drop **Sum of Sales** onto the **Y-axis**.
    b. Drag and drop the **Product** onto the **X-axis**.
    c. Drag and drop **Segment** onto the **Legend**.



MY SCREENSHOT



4. Drag the handles on the chart so that it is wide enough to fill the space under the two other visuals.



It looks like the company should continue to invest in the Paseo product and target the Small Business and Government segments.

Visual 5: Year slicer.

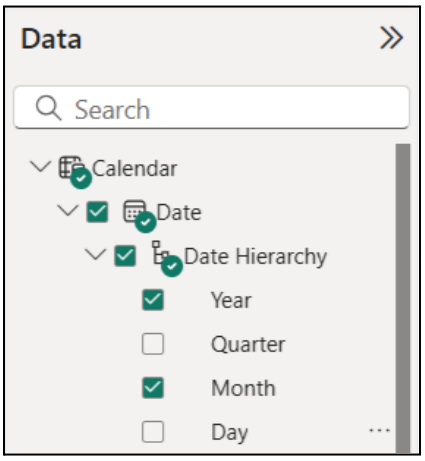
Slicers are a valuable tool for filtering the visuals on a report page to a specific selection. In this case, we can create two different slicers to narrow in on performance for each month and year. One slicer uses the Date field in the original table.

Date slicer using the original table

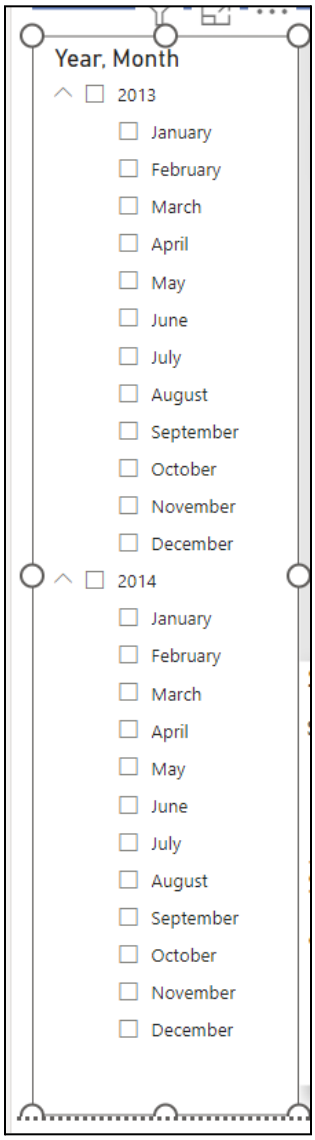
1. In the **Visualizations** pane, **choose** Slicer.
   - **Slicer** is the icon with a funnel next to a tab but without the small lightning bolt. Power BI automatically creates a numeric range slicer.

When you choose Slicer for the visual, you see the following image:

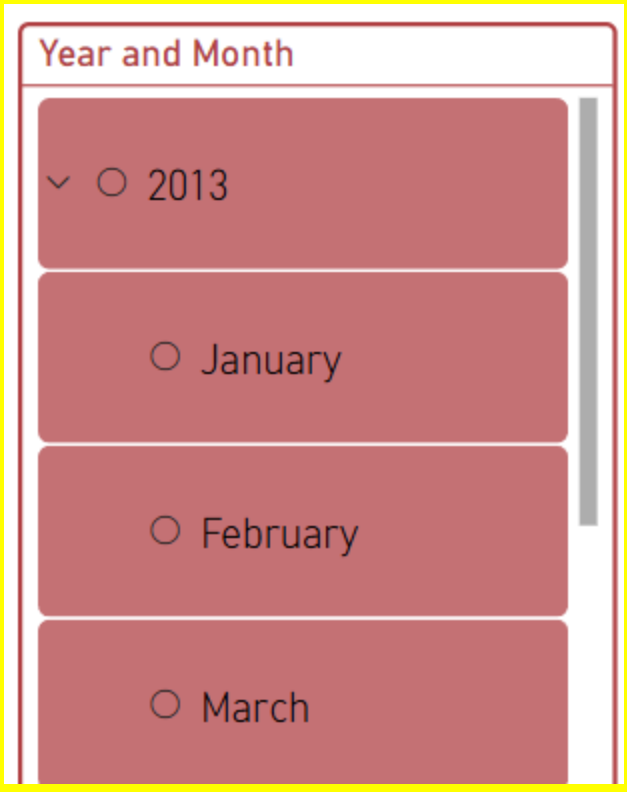
2. In the `**Data**` pane, select the `**Date**` field in the **Calendar** table. Clear the boxes for **Quarter** and **Day** so that only **Year** and **Month** are left.



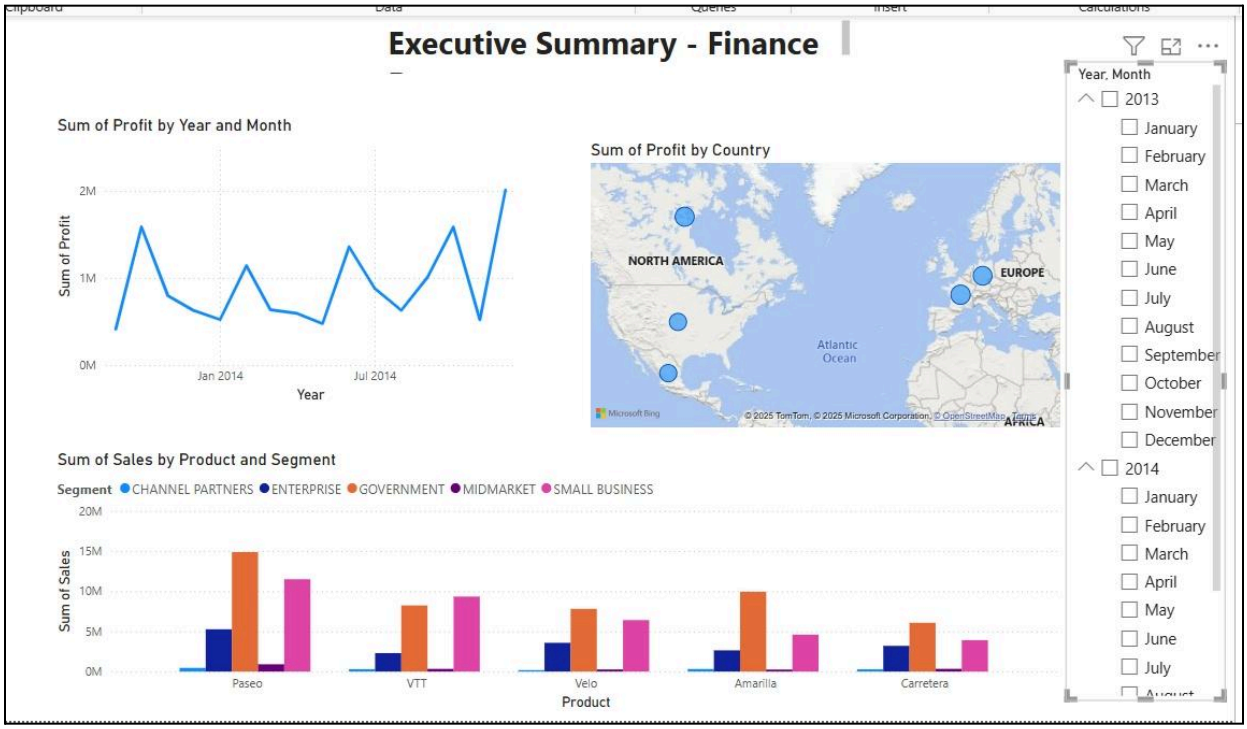
3. Expand each year and resize the visual so that all months are visible.



MY SCREENSHOT



We will use this slicer in the finished report.

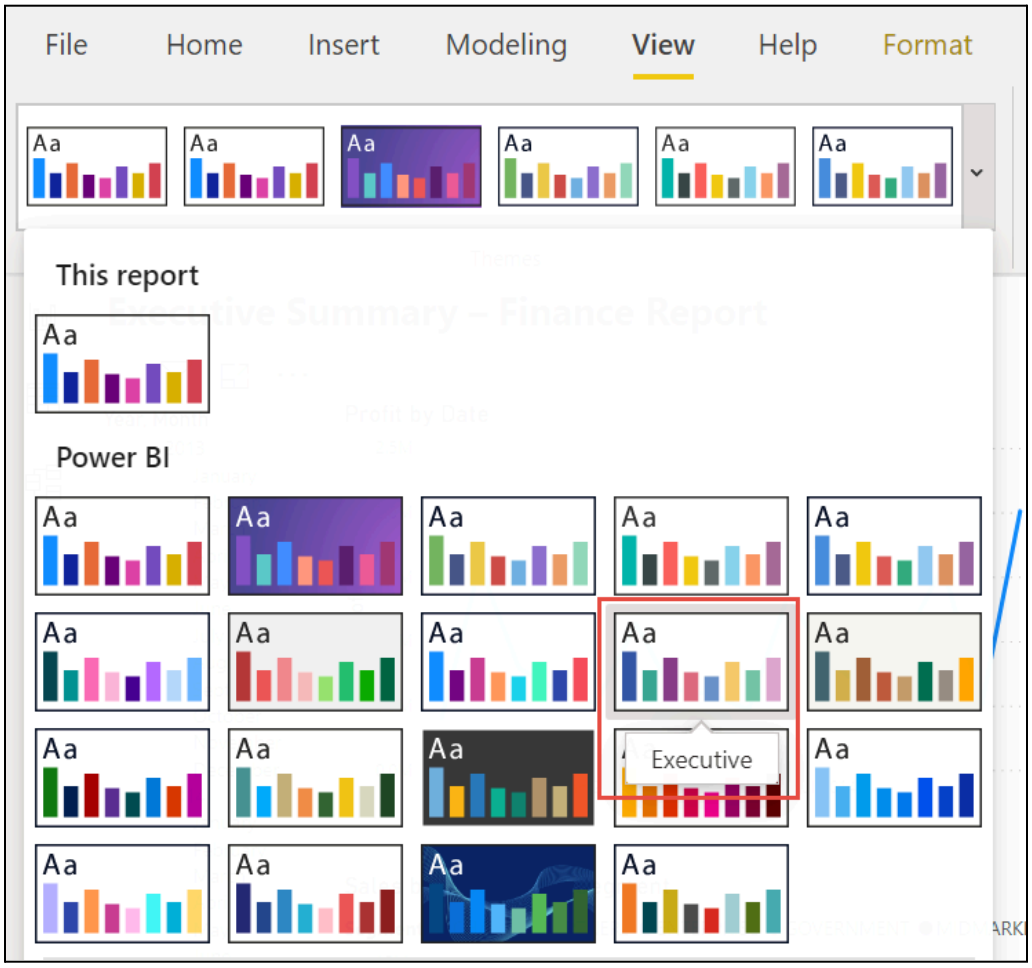


Now, if your manager asks to see just 2013 data, you can use the slicer to select that year or specific months of that year.

Step 5: Change the Theme

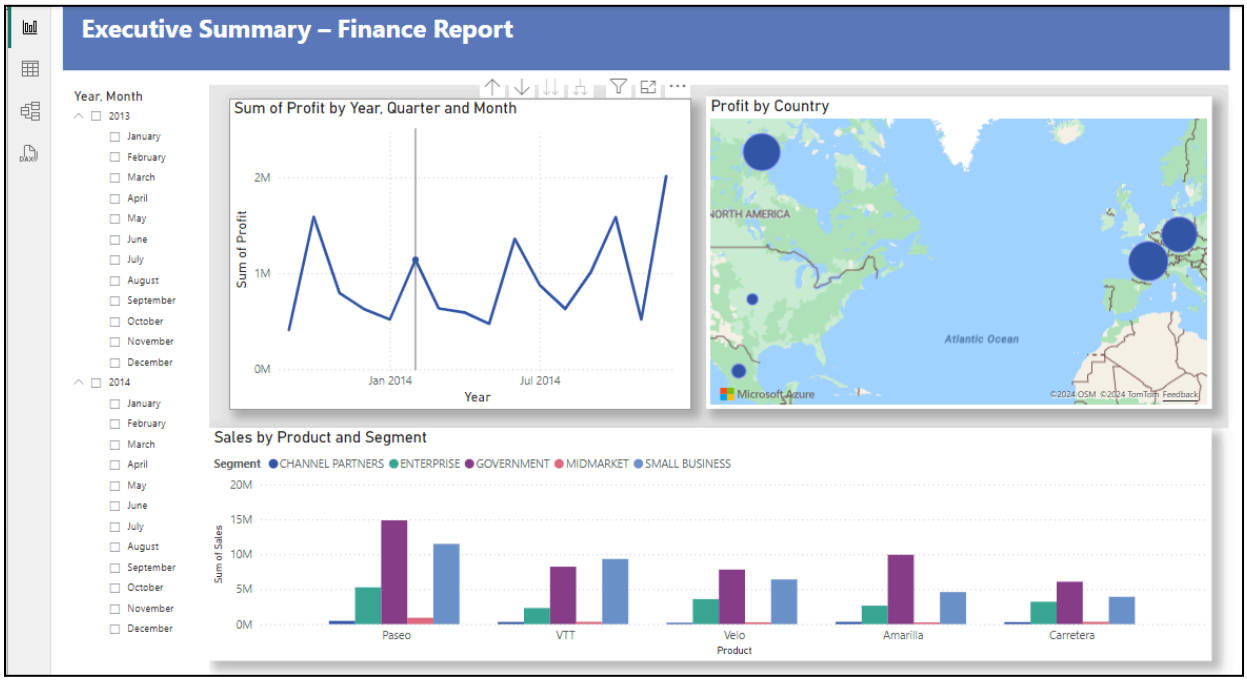
If you want to do some light formatting on this report to add more polish, here are a few easy steps.

- On the View ribbon, change the theme to Executive or any other theme.

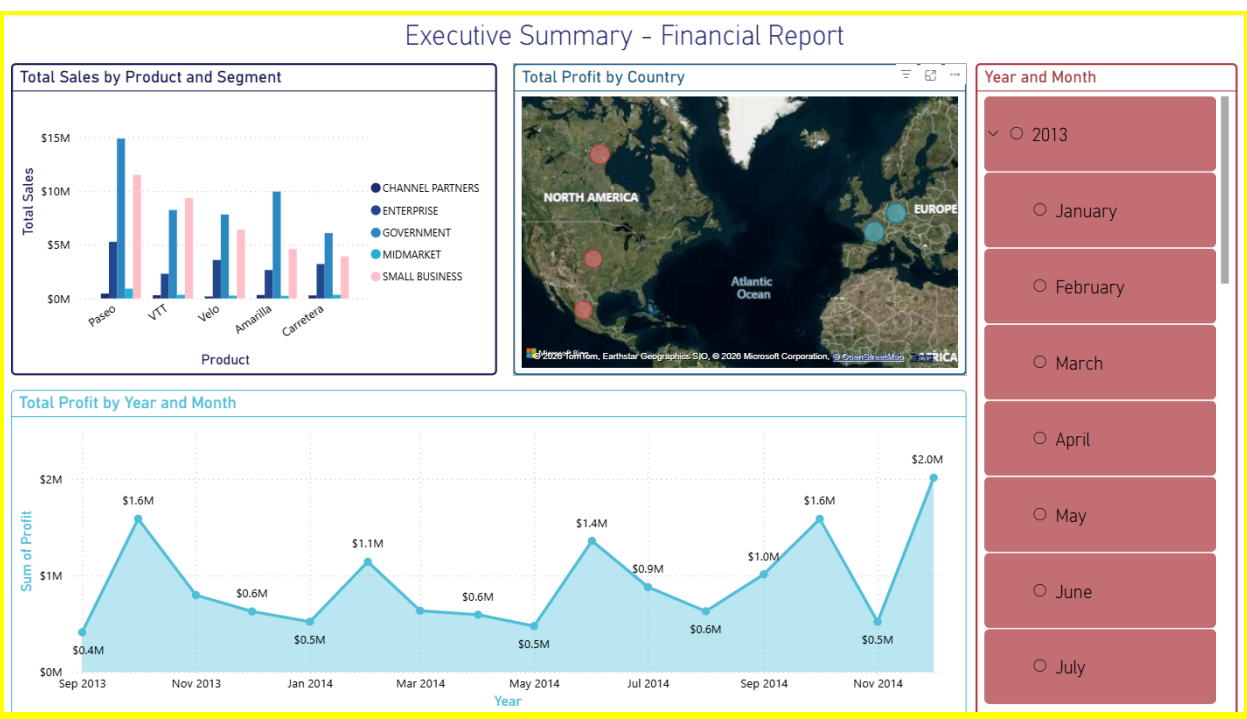


Finished report.

Here is how your final polished report will look:



MY SCREENSHOT



In summary, this report answers your manager's top questions:

- *Which month and year had the most profit?*
  December 2014.
- *Which country/region is the company seeing the most success in?*
  In Europe, specifically France and Germany.
- *Which product and segment should the company continue to invest in?*
  The company should continue to invest in the Paseo product and target the Small Business and Government segments.

Save your report.

- On the File menu, select **Save** or hold Ctrl, and then press the **S** key. If you are saving the report for the first time, you will be prompted for a name and location to save it to.

Additional Steps for Formatting the Dashboard

With the appropriate visualization selected, make the following changes on the **Visualizations** pane.

1. Select **Visual 2,** the column chart showing profit over time. Under **Format visual > General**, expand the **Title** section. Change the text in the Text field to “Profit by Month and Year” and change the Font size to 16. Expand **Effects** and toggle **Shadow** to On.

2. Select **Visual 3,** the map. Under **Format visual > Visual**, expand **Map settings and Style**, then choose Grayscale. Under **Format visual > General**, expand the Title section and change the Font size to 16. Expand **Effects** and toggle **Shadow** to On.

3. Select **Visual 4,** the clustered column chart showing sales by product and segment. Under **Format visual > General**, expand the Title section. Change the Font size to 16. Expand **Effects** and toggle **Shadow** to On.

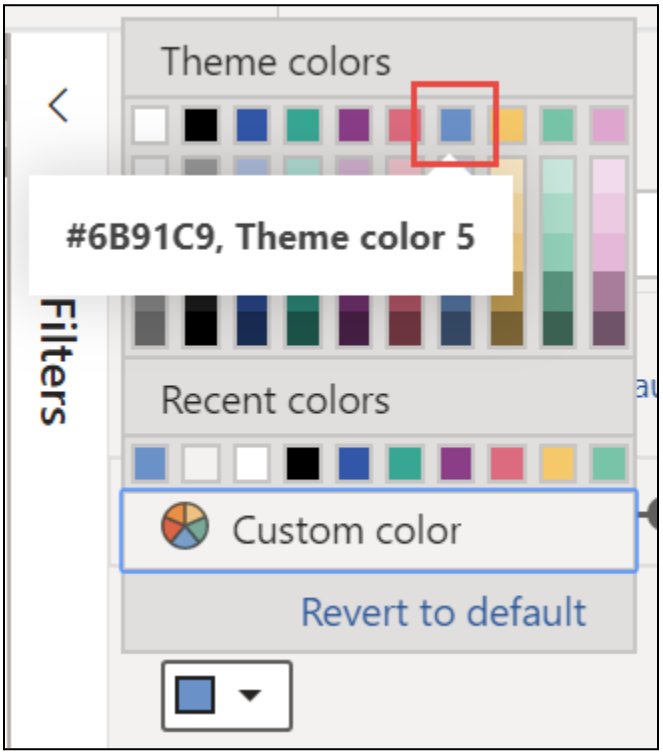
4. Select **Visual 5,** the year slicer. Under **Format visual > Visual**, expand Slicer settings > Selection and toggle the **Show "Select all"** option to On. In the Slicer header section, increase the Font size to 16.

Add a background shape for the title

1. On the Insert tab in the ribbon at the top, select **Shapes > Rectangle**. Place it at the top of the page and adjust the dimensions to be the width of the page and the height of the title.

2. In the Format shape pane, under Shape, expand the **Style** section and set **Border** to Off.

3. In the Fill section, change **Color to Theme** color 5 #6B91C9 (blue).



4. On the Format tab in the ribbon, select **Send backward > Send to back** to position the shape behind the title text.

5. Select the text in Visual 1, the title, and change the font color to White.

Add a background shape for Visuals 2 and 3

1. On the Insert tab in the ribbon at the top, select **Shapes > Rectangle**, and then adjust the dimensions to be the width and height of Visuals 2 and 3.

2. In the Format shape pane, on the **Shape** tab, expand the **Style and Border** sections and change the Transparency value to 100%.

3. In the **Fill** section, set the Color to White, 10% darker.

4. On the Format tab in the ribbon, select **Send backward > Send to back** to position the rectangle behind the visuals.

Submission:

- Take screenshots of all the examples and tasks and compile all the screenshots in a PDF document.
- Submit this PDF document through the appropriate submission link in Canvas.